Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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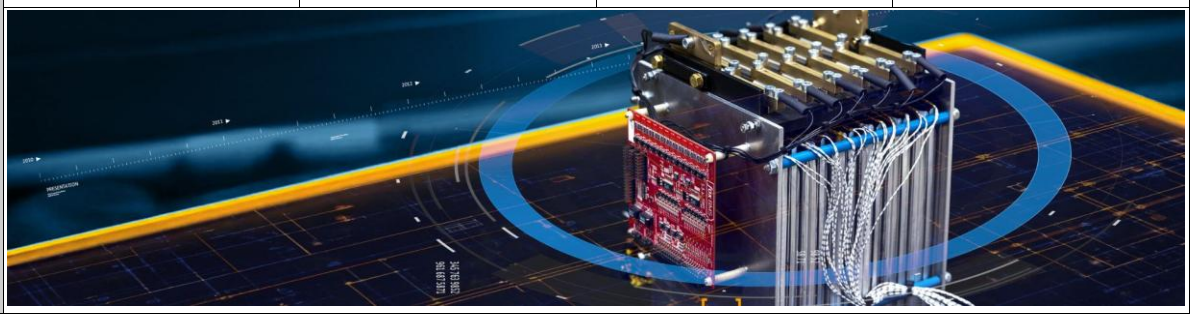
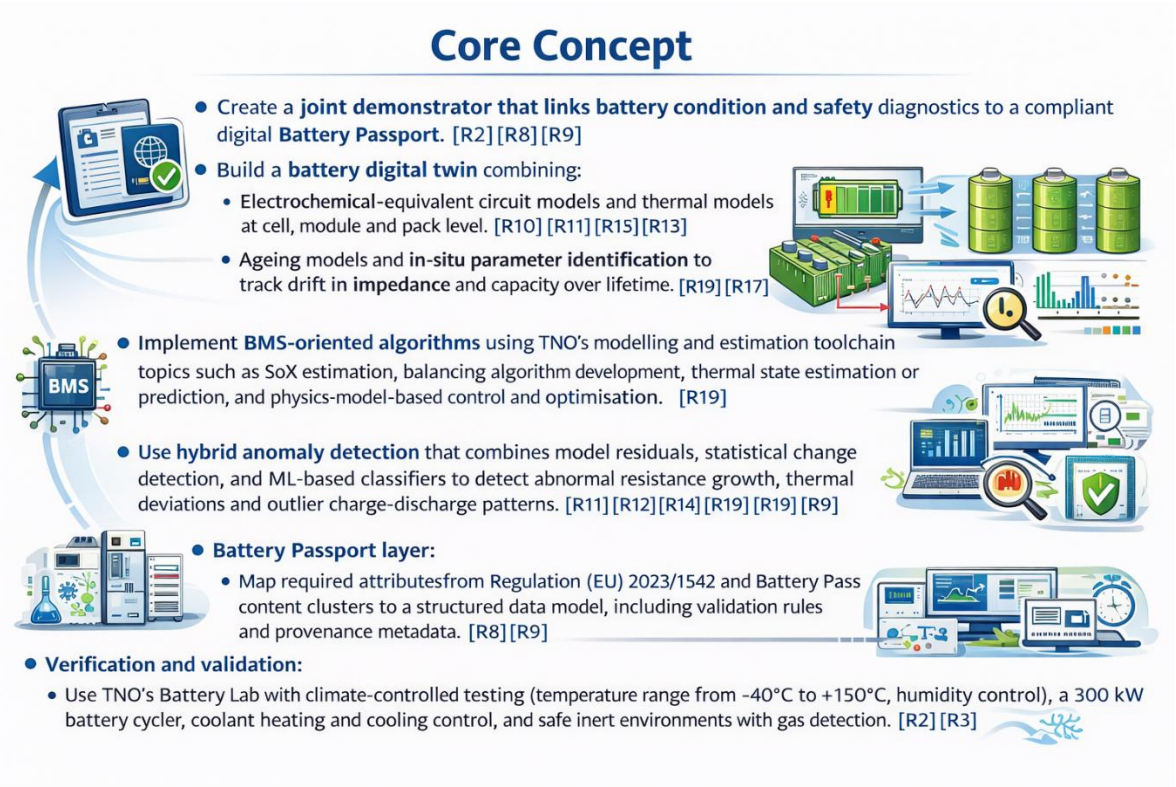
2. Technology Overview

Title of Proposed Technology	Secure Battery Passport and Early Anomaly Detection Platform using Physics-informed Battery Models and Trusted Data Spaces			
Korea's 12 National Strategic Technology Fields	Semiconductors and Display ( ), Secondary Battery (√), Advanced Mobility (√), Biotechnology ( ), Aerospace and Marine ( ), Hydrogen ( ), Cyber Security (√), Artificial Intelligence (√), Next-generation Nuclear Power ( ), Quantum ( ), Space ( ), Advanced Robotics and Manufacturing ( ) [R2][R5]			
Industrial Technology Classification [1)]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Electrical & Electronics		Information & Communications	
	Medium Category	Small Category	Medium Category	Small Category
	Batteries	Secondary Batteries	Information Security	Common Security Technologies
Concept of Proposed Technology				

Battery systems are becoming a critical enabling technology for electrified mobility, robotics, and energy-intensive applications, where safety, durability, and regulatory compliance must be managed over the full lifecycle. As batteries grow more complex and data-rich, there is increasing demand for methods that can translate raw operational data into trustworthy information about condition, ageing, and risk. At the same time, emerging regulations such as the EU Battery Passport require standardized, verifiable, and secure reporting of battery attributes throughout use. This creates a strong need for integrated approaches that combine battery modelling, diagnostics, and data governance, allowing physical battery behaviour to be consistently linked to digital representations that support safety, transparency, and compliance.



Core concept

• Create a joint demonstrator that links battery condition and safety diagnostics to a compliant digital Battery Passport. [R2][R8][R9]

• Build a battery digital twin combining:

- Electrochemical-equivalent circuit models and thermal models at cell, module and pack level. [R10][R11][R13]

- Ageing models and in-situ parameter identification to track drift in impedance and capacity over lifetime. [R19][R17]

• Implement BMS-oriented algorithms using TNO's modelling and estimation toolchain topics such as SoX estimation, balancing algorithm development, thermal state estimation or

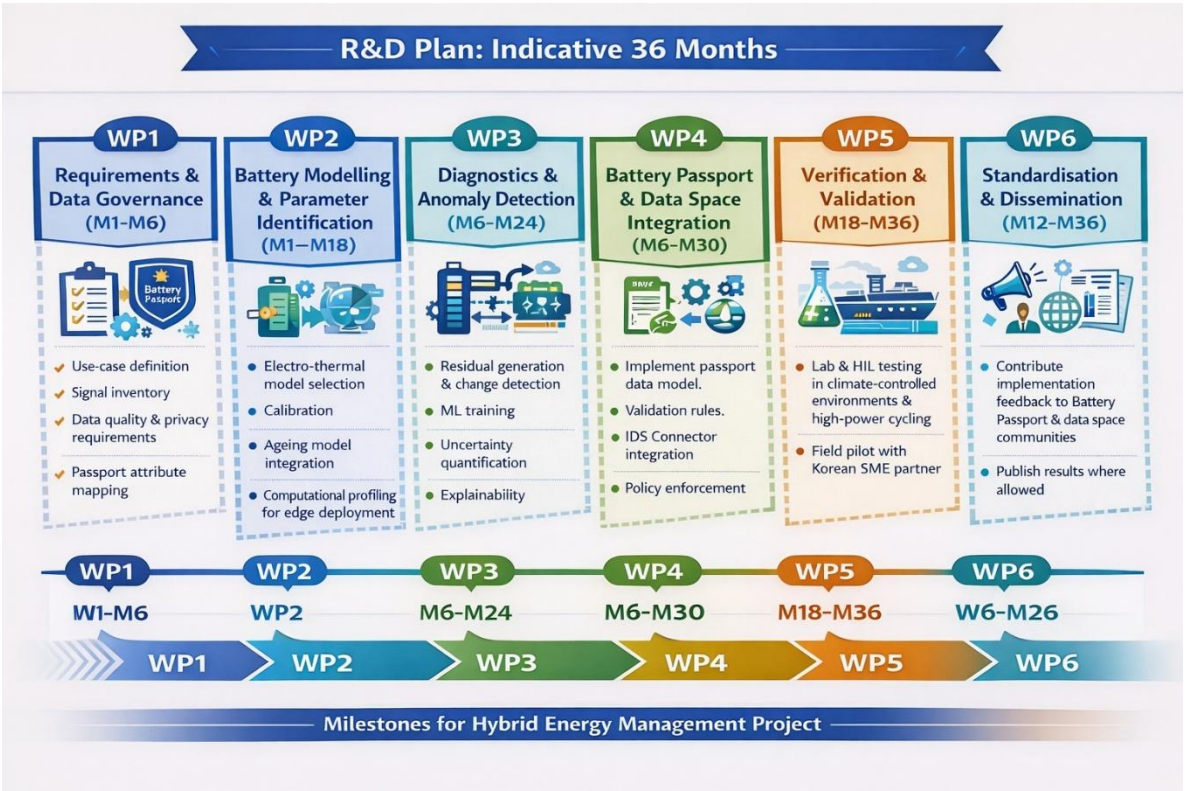
	prediction, and physics-model-based control and optimisation. [R19] • Use hybrid anomaly detection that combines model residuals, statistical change detection, and ML-based classifiers to detect abnormal resistance growth, thermal deviations and outlier charge-discharge patterns. [R11][R12][R14][R19] • Battery Passport layer: - Map required attributes from Regulation (EU) 2023/1542 and Battery Pass content clusters to a structured data model, including validation rules and provenance metadata. [R8][R9] - Export selected attributes via a secure data space interface (IDS Connector) with enforceable usage policies. [R5][R6][R7] • Verification and validation: - Use TNO's Battery Lab with climate-controlled testing (temperature range from -40°C to +150°C, humidity control), a 300 kW battery cycler, coolant heating and cooling control, and safe inert environments with gas detection. [R2][R3] - Where relevant, use Battery Lab capabilities for controlled thermal runaway testing and gas analysis to validate early-warning indicators for safety-critical scenarios. [R18]
Rationale for Joint R&D Project	Need for joint R&D • Compliance driver: Regulation (EU) 2023/1542 mandates a battery passport from 18 February 2027 for EV, LMT and industrial batteries above 2 kWh (Article 77). [R8][R9] • Data challenge: meeting passport obligations requires cross-organisation data exchange across the value chain, while preserving data sovereignty and enforceable usage restrictions. [R5][R7] • TNO capability match: TNO develops advanced battery management solutions, including SoX estimation, remaining useful life prediction (NEXTBMS) and battery passport related work. [R2][R17][R19] • Secure infrastructure: TNO's IDS Connector implementation (TNO Security Gateway) has received IDS certification and is positioned for market use in data spaces. [R6][R7] • Scientific basis: TNO has published on battery pack modelling for BMS, thermal model identification, battery capacity utilisation and charging optimisation, aligning directly with the proposed digital twin and diagnostics methods. [R10][R11][R12][R14][R15][R16]
Key Words	1. Battery Passport 2. State-of-Health (SoH) and Remaining Useful Life (RUL)

	3. Electro-thermal and ageing modelling (digital twin) 4. Hybrid anomaly detection (model residuals plus ML) 5. Trusted data sharing (IDS Connector, data sovereignty) [R5][R8][R9][R14]

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	3	End of the Project	6
R&D Objectives	R&D Objectives 1) Battery digital twin: develop multi-fidelity electro-thermal and ageing models suitable for embedded or edge execution, with calibration workflows and in-situ parameter identification. [R11][R13][R19] 2) SoX and RUL: implement estimators and predictors that fuse physics models with operational data and output calibrated uncertainty, with verification against lab reference tests. [R2][R17][R19] 3) Anomaly detection: design a hybrid pipeline (residual diagnostics, change detection, ML) for early warning, root-cause hypotheses and safety risk scoring. [R11][R12][R14][R19] 4) Battery Passport layer: implement a structured data model aligned to Regulation (EU) 2023/1542 and Battery Pass content clusters, plus automated validation and provenance tracking. [R8][R9] 5) Trusted data exchange: deploy an IDS-based architecture (data space connector, identity, policy enforcement, logging) using TNO Security Gateway where applicable. [R5][R6][R7] 6) Validation: demonstrate performance in TNO Battery Lab and partner-relevant field trials, including climate-controlled cycling and safety testing where required. [R2][R3][R18]			
R&D Target Specifications	R&D Target Specifications (project targets) • SoC estimation accuracy: target median absolute error below 2 percentage points on validation cycles. • SoH estimation: target capacity estimate error below 5 percent relative to reference lab tests. • RUL prediction: target error within 10 percent for ageing trajectories represented in the training and validation data. • Anomaly detection: target false alarm rate below 1 per 1000 operating hours and detection lead time of at least 10 minutes for selected fault scenarios. • Data exchange: target automated passport export through an IDS Connector with enforceable usage policies and auditable logs. (Targets are defined for project planning and will be refined in WP1 based on partner requirements.) [R5][R8][R9]			

R&D Plan



R&D Plan (indicative 36 months)

WP1 Requirements and data governance (Months 1 to 6): use-case definition, signal inventory, data quality and privacy requirements, and passport attribute mapping.

WP2 Battery modelling and parameter identification (Months 1 to 18): electro-thermal model selection, calibration, ageing model integration, and computational profiling for edge deployment.

WP3 Diagnostics and anomaly detection (Months 6 to 24): residual generation, change detection, ML training, uncertainty quantification, and explainability.

WP4 Battery Passport and data space integration (Months 6 to 30): implement passport data model, validation rules, IDS Connector integration and policy enforcement.

WP5 Verification and validation (Months 18 to 36): lab and HIL testing in climate-controlled environments and high-power cycling, followed by a field pilot with the Korean SME partner.

WP6 Standardisation and dissemination (Months 12 to 36): contribute implementation feedback to Battery Passport and data space communities; publish results where allowed.

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	TNO (Research Institute) • Provide modelling and algorithm development for SoX, RUL and anomaly detection. • Provide access to Battery Lab methods, characterisation workflows and HIL test approaches. • Design and implement the trusted data exchange architecture using IDS Connector technology. • Lead technical integration, verification and reporting to TIPA.
Expected R&R of Korean SME Partner	Korean SME Partner • Provide the application context, operational constraints and functional requirements. • Provide representative data streams and support instrumentation or BMS data access within agreed governance. • Host and operate the prototype in a relevant environment for field validation. • Contribute to exploitation planning, productisation and market access (Korea and EU as applicable).

5. Expected Outcomes/Impact

Expected Outcomes and Impact	Technical outputs • Battery digital twin library (electro-thermal plus ageing) and calibration toolchain for partner-specific configurations. • Hybrid diagnostics and anomaly detection software (edge plus cloud), including uncertainty estimates and explainability artefacts. • Battery Passport export service that selects, validates and publishes required attributes. • Reference implementation of trusted data exchange using IDS Connector components and policies. Validation • Lab and HIL validated performance, including testing in climate-controlled environments and high-power cycling setups. • Field pilot results demonstrating operational feasibility and integration guidance. Impact • Reduced safety risk and downtime through earlier anomaly detection. • Accelerated compliance readiness for Battery Passport obligations from 2027 for relevant battery categories. • Improved trust and willingness to share data across the value chain through enforceable data sovereignty controls.

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☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization